

for SR, as seen in (18) above, the lengthening at the end of ϕ caused by the application of FL has the effect of introducing enough distance between the two primary word stresses to create well-formed rhythmical alternations of prominence. We will return to this point in Chapter 10 with reference to poetic meter.

6.4. *The phonological phrase in languages other than Italian*

The definition of the ϕ domain proposed in (5) above makes reference to very general syntactic notions, such as those of phrasal head and recursive side of a phrase with respect to the head. They thus make predictions about the ϕ domain not only for Italian but for all configurational languages, that is, languages whose syntactic structure can be represented within the $\bar{X}$ system. In this section, we will show that the phonological phrase proposed above is indeed the domain of application of a number of phonological rules in a variety of languages. As far as the ϕ restructuring rule is concerned, it will be shown that in addition to languages such as Italian where it is optional, there are languages in which restructuring is obligatory (see Hayes, to appear) and still other languages in which it is not an option at all (see also Nespor and Vogel, 1982).

6.4.1. *Other right recursive languages*

The first language we will consider is English. We will show that the phonological phrase is the domain of application of two phonological rules: Iambic Reversal (see Liberman and Prince, 1977) and the Monosyllable Rule (see Selkirk, 1972). This proposal is not new; in fact, ϕ was proposed as the domain of application of these rules by Selkirk (1978b). Since the domain of Selkirk's ϕ is somewhat different from ours, however, we will show that it is, in fact, our definition of ϕ that provides the appropriate description of the environments in which the two rules apply. In particular, we will demonstrate that the phonological phrase in English has the same domain as it has in Italian, as is to be expected given that both languages are right recursive. It will be shown, further, that restructuring of ϕ is optional in English, as in Italian, and that the rules given above are thus the correct rules for defining ϕ .

Iambic Reversal (IR) is a rule that has essentially the same effect as Stress Retraction in Italian, as is shown in (21). (See Liberman and Prince, 1977, for details of this phenomenon.)

- (21) a. thirtéén mén → thírteen mén
 b. Tènnessé aír → Ténnessee aír

Like SR, this rule does not apply in just any sequence of two words that

contains a stress clash. Liberman and Prince claim that the rule is optional and that 'the acceptability of a stress clash depends, in part, on the extent to which the interstress interval is prone to be lengthened, independently, by prepausal lengthening' (p.320). According to the analysis presented here, IR is not optional as to its environment, but rather is a ϕ internal rule, as is shown in (22)-(24).

- (22) a. More than fifteen *cárpenters* are working in the house.
(< *fifteén*)
b. The *kángaroo's* life is full of surprises. (< *kangaroó*)
- (23) a. John *pérseveres* gládlly. (< *persevéres*)
b. Given the chance, rabbits *réproduce* quíckly. (< *reprodúce*)
- (24) a. John *persevéres* gládlly and diligently. (* *pérseveres*)
b. Given the chance, rabbits *reproduce* véry quickly. (* *réproduce*)

The sentences in (22) exemplify the application of IR within ϕ as predicted by the definition of ϕ in (5); those in (23) show that IR applies within ϕ restructured according to the rule in (13); and those in (24) show that IR does not apply outside the ϕ domain. The examples in (23) and (24), moreover, offer additional evidence for the observation that the nonbranchingness of the first complement to the right of the head is the appropriate criterion for ϕ restructuring. This is also the crucial difference between the phonological phrase proposed by Selkirk (1978b) and that proposed here, since in Selkirk's proposal the phonological phrase may not extend to include any constituent to the right of the head of a phrase. Since there are, in fact, cases in which this is necessary, the present analysis must be preferred on empirical grounds.

It should be observed that in English, as in Italian, ϕ is also the domain of Final Lengthening. Liberman and Prince's observation that IR does not apply if the interval between the two stresses is lengthened is thus easily accounted for: since lengthening takes place at the end of ϕ , and since both words involved in IR must belong to the same ϕ , it is clear that the one phenomenon excludes the other.

Another rule of English that offers evidence for the ϕ level of prosodic structure is the rule that reduces monosyllabic words that do not belong to any of the lexical categories N, V, or A. As is demonstrated by Selkirk (1978b), this Monosyllable Rule applies only if the monosyllable is labeled weak with respect to another syllable in the same ϕ that is labeled strong. The fact that the rule applies in (25a) but not in (25b), which represent Selkirk's (1978b) examples with our division into constituents, is thus accounted for.

- (25) a. [The sluggers]_φ [boxed]_φ [in the crowd]_φ (reduced *in*)
 b. [The cops]_φ [boxed in]_φ [the crowd]_φ (unreduced *in*)

In the first example, the preposition *in* is the leftmost node of a ϕ and is thus weak with respect to the strong node dominating *crowd*. Hence, *in* may undergo reduction. In the second example, on the other hand, *in* may not be reduced, since it is the rightmost node of the ϕ containing *boxed in*, and is thus labeled strong with respect to its sister, which is weak.

The second language we will discuss here in relation to the phonological phrase is French, a language in which the restructuring of ϕ is not an available option. That is, ϕ may only include a head X and whatever is on its nonrecursive side until another head outside of the maximal projection of X is reached. The rule we will consider is Liaison, a rule that applies in all styles of speech, but whose contexts of application become larger as the degree of formality increases (see Selkirk, 1972). We will restrict our attention to the colloquial style of speech, since it is only in this style that Liaison applies in a purely phonological environment. As has been shown in Morin and Kaye (1982), the contexts of application of Liaison in elevated styles of speech are partly morphologically determined. In colloquial French, however, the purely phonological Liaison rule applies within the domain of the phonological phrase, as exemplified in (26).

- (26) a. Cette famille a [trois beaux_enfants]_φ
 ‘This family has three beautiful children.’
 b. Les enfants [sont_allés]_φ à l’école.
 ‘The children went to school.’

The examples in (27) show that Liaison does not apply between words that belong to two different ϕ s, and those in (28) show that it does not apply between the head of a phrase and the first nonbranching complement on the recursive side. The latter set of examples thus demonstrates that the restructuring of ϕ is not a property of French.

- (27) a. Jean a [des livres]_φ // [assez nouveaux]_φ
 ‘Jean has some rather new books.’
 b. Nos invités [sont arrivés]_φ // [en retard]_φ
 ‘Our guests arrived late.’
- (28) a. [Les maisons]_φ // [italiennes]_φ coûtent beaucoup.
 ‘Italian houses are expensive.’
 b. Le garçon [les aidait]_φ // [activement]_φ
 ‘The boy helped them actively.’

